S5 - Glossary

Shared terminology across articles, with RVC/sleep-specific entries

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Abstract This document defines the core terminology used across articles in this repository, with additional entries specific to the RVC_SLEEP spindle-geometry study (two-dataset A2r timing result in ANPHY/BOAS; RichSleep same-estimand timing test inconclusive). Entries cover the theoretical framework, the computational pipeline, embodied anchor metrics, the family of primary metric scalar summaries, the subject-weighted statistical protocol, and the robustness-audit methods used in this article. Definitions reflect the current implementation as of the RVC_SLEEP analysis freeze (2026-07-03/04) and the 2026-08-30 claim-scope revision; version-specific changes to metric computation are noted in the relevant entry.

Note: This is a living document. Definitions are updated to match the current pipeline version. Version-specific changes to metric computation are noted in the relevant entry.

Reading guide for reviewers. *Where relevant, entries distinguish three layers: (1)* ***Implemented*** *- the exact computation in the pipeline; (2)* ***Interpretation*** *- the most defensible empirical reading; (3)* ***Speculative / Not established*** *- theoretical mappings not yet directly confirmed.*

A - D

A1r / A2r - The two primary subject-weighted contrast estimands used in this article. **A1r**: event-vs-matched-control specificity (`event_bin - matched_control_median`, subject-weighted). **A2r**: event-vs-pre_far within-spindle timing (`event_bin - mean(pre_far bins, same event)`, subject-weighted). Adopted as the corrected primary estimand (over the earlier event-weighted A1/A2) because of BOAS's uneven per-subject spindle-event counts.

Anchor-augmented surface - An alternate MNPS coordinate-export configuration (ANPHY/BOAS only) in which the `e_m` sub-coordinate's EEG term is partially or fully replaced by the dataset's primary body-derived anchor signal. Used only for the Pass B surface-sensitivity construction-validity check; never used as the primary geometric claim (dual-surface contract).

Attractor - A stable region in the neural state manifold to which trajectories tend to return after small perturbations. Geometrically, a local minimum in the potential landscape. **Speculative / Not established** in the context of this article: no attractor-landscape reconstruction was performed; the term is used only in the general NDT background sense.

Cardioresp_anchor_index - RichSleep's primary Embodied Anchor Principle (EAP) composite index: a dual-arm anchor built from **simultaneous** ECG and respiratory channels with near-100% subject coverage. Used as the primary anchor in the RichSleep EAP-moderation test (R6), specifically because it removes the single-channel-missingness limitation of ANPHY's cardiac-only and BOAS's respiratory-only anchors.

Coordinate (`m`, `d`, `e`) - The canonical 3D MNPS chart. `m` (metastability/mobility): how variable/mobile the state trajectory is. `d` (deviation): distance from an optimal integration-segregation balance point. `e` (entropy/entropic energy): disorder/complexity of the underlying signal. **Interpretation**: higher `d` during a spindle event reflects the state moving further from that balance point; this article does not interpret this directionally as "better" or "worse."

`d_n`, `d_l`, `d_s` - Three stratified sub-coordinates recomposed into `d`. **Interpretation**: `d_n` as network-integration deviation (this article's most reproducible tested endpoint in ANPHY/BOAS A2r); `d_l` as local clustering-based segregation load; `d_s` as representational-geometry deviation. **Implemented**: the NMD `mnps_9d.subcoords` feature map in the hashed run configs, not a schematic heat-kernel formula (S1.2).

Dual-surface contract - The rule that the primary MNPS geometry export never contains body-signal-derived features (`primary_eeg_only`), and that any body-signal-augmented export (`anchor_aug`) is used strictly for a construction-validity check (Pass B), never for the primary specificity/timing claim. Central to interpreting Pass B correctly (S1.4, main text R5).

E - L

Embodied Anchor Principle (EAP) - The NDT proposition that bodily/interoceptive rhythms (cardiac, respiratory, ocular) can condition the same latent landscape that governs cortical state-space dynamics. In this article, tested strictly as a **moderator** of the EEG-only spindle-locked geometry effect (Pass C / RichSleep R6), not as a component of the geometry itself. **Not supported** under the tonic composite anchors and underpowered phase-resolved tests reported here (main text R5–R6b); not a general rejection of body-brain coupling.

`e_e`, `e_s`, `e_m` - Three stratified sub-coordinates recomposed into `e`. **Implemented in this contract**: `e_s = 0.5 · eeg_alpha_theta + 0.5 · eeg_gamma`; primary-surface `e_m = 0.6 · eeg_hjorth_mobility + 0.4 · eeg_beta_alpha`. `e_s` is not permutation entropy here. `e_m` is the one endpoint whose feature composition differs on the anchor-augmented surface (Dual-surface contract). See S1.2.

Event-locked table - The per-(subject, window, condition, bin_label) analysis-ready table produced by the NMD pipeline, aligning each detected sleep spindle against five peri-event bins (`pre_far`, `pre_near`, `event`, `post_near`, `post_far`) and a matched-control pool of non-spindle N2 windows. Control **rows** are 30 s MNPS-grid expansions; analysis median-collapses them per `matched_event_id` (S1.1). Not 1–3 independent control rows per spindle.

M - R

Matched control - A non-spindle N2 window from the same subject and recording session, drawn at least 30 s from any spindle/K-complex/arousal, used as the comparison condition for the A1r specificity contrast.

Meta-Noetic Phase Space (MNPS) - The low-dimensional geometric chart (3D: m, d, e; 9D stratified: $m_a, m_e, m_o, d_n, d_l, d_s, e_e, e_s, e_m$) that NDT uses to project raw electrophysiology into an auditable, release-fixed state-space representation. See S1.2.

NeuralManifoldDynamics (NMD) - The versioned processing pipeline that ingests raw EDF/PSG recordings and exports per-epoch MNPS coordinates, Jacobian tensors, anchor-state indices, and event-locked analysis tables. Every claim in this article is read against the NMD `summarize` output for the relevant run (config-hash-verified), not against raw signals directly.

Pass A / A1 / A2 / A1r / A2r / decomposition - The frozen analysis sequence for ANPHY/BOAS: Pass A (base N2/coverage filter); A1 (event-weighted, pre-amendment specificity); A2 (event-weighted, pre-amendment timing); A1r/A2r (corrected, subject-weighted versions of the same two contrasts); A1 decomposition (splitting the A1r delta into a timing component and a baseline-separation component, resolving the `d_n` cross-dataset sign conflict - main text R4).

Pass B - The surface-sensitivity check: does the anchor-augmented surface change the A1r/A2r subject-weighted deltas relative to the primary surface? A construction-validity check, **not** an EAP-moderation test (main text R5).

Pass C - The EAP-moderation test itself: `delta ~ anchor_pre_far + nuisance covariates`, weighted least squares with cluster-robust SE by subject, plus a per-subject-slope sign-flip permutation sensitivity check. Run on ANPHY (cardiac-only anchor) and BOAS (respiratory-only anchor) in the frozen analysis; re-run on RichSleep (dual-arm cardiorespiratory anchor) as an out-of-sample verification (main text R6).

Primary metric (endpoint) family - The frozen 9-endpoint set $m, d, e, d_n, d_l, d_s, e_e, e_s, e_m$ used for every specificity/timing/moderation test in this article. Reachability-cone and Meta-Noetic Jacobian (MNJ) endpoints were planned as secondary layers but were not run under the current analysis freeze.

Rhythmic Variance Control (RVC) - The NDT hypothesis under test: that rhythm-defined events (here, sleep spindles) act as transient "variance-control" or gating-precision events, producing a measurable, reproducible reconfiguration of local state-space geometry. **Internal validated result** for the ANPHY+BOAS `d_n` timing signature specifically (main text R3); RichSleep A2r of the same estimand is **inconclusive**. Treating that signature as a general RVC / gating-precision **mechanism** is a **speculative extension**.

S - Z

Sign-flip permutation test - The primary hypothesis test used throughout this article: randomly flips the sign of each subject's mean delta $n = 2000$ times and compares the observed grand mean against the resulting null

distribution. Chosen over a parametric test because it directly tests the within-subject direction question without a distributional assumption; one-sample t-test and Wilcoxon signed-rank on the same subject means are reported as sensitivity checks, not as the primary p-value.

Statistical method (BH-FDR) - Benjamini-Hochberg false discovery rate correction ((Benjamini & Hochberg, 1995)), applied fresh within each frozen test family in this article (18 tests for ANPHY+BOAS A1r/A2r; 9 tests for each RichSleep A1r/A2r family; 54 tests for Pass C; 126 fitted rows for the RichSleep EAP family). Threshold: $q < 0.05$ throughout. Composites m, d, e are not independent of the stratified coordinates; FDR still treats them as separate tests.

Subject-weighted estimand - The correct unit of analysis for this article's uneven-event-count datasets: every event-level delta is first collapsed to one subject-mean delta before any test statistic is computed, so that subjects with many spindle events do not numerically dominate subjects with few.

Thalamic Reticular Nucleus - Variance Gating (TRN-VG) - A proxy interpretation, not a direct measurement: spindle/slow-oscillation events are treated as thalamocortical-gating **proxies** in this line of work, because none of the three PSG datasets can directly observe TRN activity. This article makes no claim beyond "TRN-VG-consistent," and does not claim TRN-VG is validated.

YASA - Yet Another Spindle Algorithm ((Vallat & Walker, 2021)), the open-source automated sleep-staging and spindle-detection tool used to detect all spindle events in this article (C3-A2 channel, sigma 12-15 Hz, N2-only).

Bibliography

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